

Experiment No. 7:**To determine the time constant of an RC circuit.****7.1 Objectives**

The main objective of this lab is to construct an RC circuit and to determine the time constant of that RC circuit.

7.2 Prelab

Students must have clear idea about the objectives and related theory of the experiment. Lab Demonstrator will confirm that the students have submitted their previous group lab reports in printed form before giving them attendance. The marks of performance depend greatly on this part.

7.3 Outcomes

After completing this lab work, students will be able to answer the following questions:

- What is capacitor and capacitance? How a capacitor can be charged and discharged in a circuit.
- What is the significant of time constant in an RC circuit?
- How the time constant of an RC circuit can be obtained?
- How the voltage across the capacitor in an RC circuit varies with time for charging and discharging of the capacitor?

7.4 Timing and Length of Investigation (Total 3 Hours)

- **Lab Preparation (15 minutes):**
 - For attendance and equipment distribution among the groups.
- **Lecture on Theory (40 minutes):**
 - The teacher will clarify the objective and theory of the experiment.
- **Lecture on Procedure 30 minutes):**
 - Demonstrator will make the students understand the procedure of the experiment.
- **Experimental Work (80 minutes):**
 - Design of experiment: 10 minutes
 - Data collection: 30 minutes
 - Data analysis, drawing graphs and completing the result part: 40 minutes.
- **Post Lab Discussion (15 minutes):**
 - The teacher will summarize the total lab work and have a discussion with the students related to the questions given in the outcomes part.
- **Report Submission:**
 - Students will prepare their lab report and submit it on the next class.

7.5 Theory

Capacitors are used in timing circuit in many devices. The time that the dome lights inside a car stay on after turning off the cars ignition at night is one example of how a capacitor can be used to maintain the lighting long enough to remove the key and collect things before exiting. The values we use to characterize these kinds of circuits is given by the time constant defined as: $\tau = RC$, where R is the circuit resistance and C is the capacitance. In this lab, we will observe the charging and discharging of a capacitor and determine the time constant of a RC circuit.

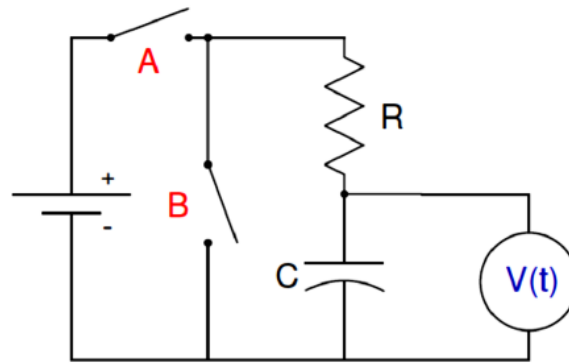


Figure 7.1: Circuit for RC charge-discharge measurement where $V(t)$ is the potential difference across the capacitor as a function of time.

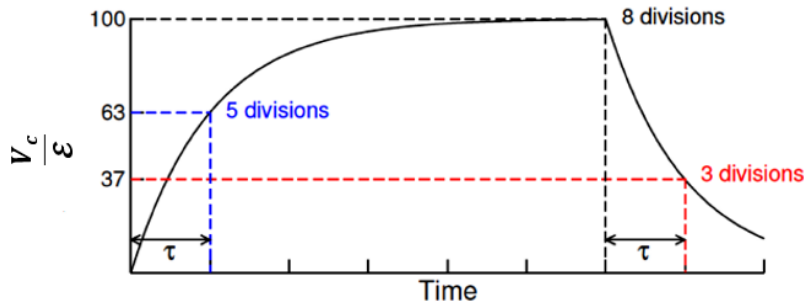


Figure 7.2: Potential difference across a capacitor in an RC circuit as a function of time.

The time constant can be determined by observing the either the charging or the discharging process of the capacitor as the Fig. 7.2 shows. For the charging process, τ is the time for $V(t)$ to reach 63% of its final value. For the discharging process, τ is the time for $V(t)$ to fall 63% from its initial value.

Charging a Capacitor:

The capacitor of capacitance C in Fig. 7.1 is initially uncharged. To charge it, we close switch on in point A. This completes an RC series circuit consisting of the capacitor, an ideal battery of emf $\mathcal{E}$, and a resistance R . Charge begins to flow (current exists) between a capacitor plate and a battery terminal on each side of the capacitor. These current increases the charge q on the plates and the potential difference $V_c (=q/C)$ across the capacitor. When that potential difference equals the potential difference across the battery (which here is equal to the emf $\mathcal{E}$), the current is zero. From Eq. ($q = CV$), the equilibrium (final) charge on the fully charged capacitor is equal to $C\mathcal{E}$. To examine the charging process, we begin by applying the loop rule to the circuit, traversing it clockwise from the negative terminal of the battery.

We find from Kirchhoff's loop theorem:

$$\mathcal{E} - iR - \frac{q}{C} = 0$$

$$\mathcal{E} = iR + \frac{q}{C} \quad [i \text{ \& } q = \text{two variables, } i = \frac{dq}{dt}]$$

$$\mathcal{E} = R \frac{dq}{dt} + \frac{q}{C} \dots\dots\dots(1)$$

$$\text{And,} \quad q = q_0(1 - e^{-\frac{t}{RC}})$$

$$CV_c = C\mathcal{E}(1 - e^{-\frac{t}{RC}}) \quad \text{since, } [q(t) = CV_c(t)] \text{ \& } [q_0 = C\mathcal{E}]$$

$$\ln(1 - \frac{V_c}{\mathcal{E}}) = (-\frac{1}{RC})t \dots\dots\dots(2)$$

Discharging a capacitor:

Assume now that the capacitor of Fig. 7.1 is fully charged to a potential V_0 equal to the emf $\mathcal{E}$ of the battery. At a new time, $t = 0$, switch is thrown from A to B so that the capacitor can discharge through resistance R . Now the charge $q(t)$ on the capacitor and the current $i(t)$ through the discharge loop of capacitor and resistance now vary with time.

With no battery in the discharge loop, $\mathcal{E} = 0$, Eq. (1) becomes using Kirchhoff's loop theorem:

$$iR + \frac{q}{C} = 0$$

$$R \frac{dq}{dt} + \frac{q}{C} = 0 \quad \left[i = \frac{dq}{dt} \right]$$

And, $q = C\mathcal{E}e^{\frac{-t}{RC}}$
 $CV_c = C\mathcal{E}e^{\frac{-t}{RC}} \quad \text{since, } [q(t) = CV_c(t)] \text{ \& } [q_0 = C\mathcal{E}]$
 $\ln\left(\frac{V_c}{\mathcal{E}}\right) = \left(-\frac{1}{RC}\right)t \dots\dots\dots(3)$

Comparing Eq. 2 and 3 with $y = mx+c$ and plotting a graph of " $\ln\left(1 - \frac{V_c}{\mathcal{E}}\right)$ vs t " for charging and " $\ln\left(\frac{V_c}{\mathcal{E}}\right)$ vs t " for discharging, we get the value of τ as $\tau = -\frac{1}{m}$, where m is the slope of the graphs.

7.6 Apparatus

Power supply, circuit board, resistor, capacitor, multi meter, stopwatch and connecting wires.

7.7 Procedure

- Construct an RC circuit on the circuit board as the circuit diagram shows.
- Applying a sufficient voltage from the power supply, observe the charging of the capacitor and note the voltage differences across the capacitor with time.
- Disconnect the power supply from the circuit, observe the discharging of the capacitor with time. Also note the voltage differences across the capacitor with time.

7.8 Experimental Data

Table 7.1: Charging & Discharging of an RC circuit.

Maximum potential difference ($\mathcal{E}$) =

Time (s)	Charging capacitor		Discharging capacitor	
	V_c (volt)	$\ln(1 - \frac{V_c}{\mathcal{E}})$	V_c (volt)	$\ln(\frac{V_c}{\mathcal{E}})$
0				
30				
60				
90				
120				
150				
180				
210				
240				
270				
300				
330				
360				
390				
420				
450				
480				
510				
540				
570				
600				
630				
660				
690				

7.9 Analysis and Calculation

- Use EXCEL to plot “ V_c **versus** t ” graphs for charging and discharging.
- From the relation: $\tau = RC$, estimate the value of time constant.
- For charging, plot a graph of “ $\ln(1 - \frac{V_c}{\mathcal{E}})$ **vs** t ” and from the value of slope calculate the value of time constant.
- For discharging, plot a graph of “ $\ln(\frac{V_c}{\mathcal{E}})$ **vs** t ” and from the value of slope again calculate the value of time constant.

Calculation: Charging a capacitor

Theoretically:

Time constant, $\tau = RC =$

Experimentally:

From the graph: Slope, $m = -\frac{1}{\tau}$

$$\tau = -\frac{1}{m}$$

Error of $\tau = \frac{\tau_{th} - \tau_{ex}}{\tau_{th}} \times 100\%$

Calculation: Discharging a capacitor

Theoretically:

Time constant, $\tau = RC =$

Experimentally:

From the graph: Slope, $m = -\frac{1}{\tau}$

$$\tau = -\frac{1}{m}$$

Error of $\tau = \frac{\tau_{th} - \tau_{ex}}{\tau_{th}} \times 100\%$

Table 7.2: Values of time constant, τ .

From the graphs		Theoretically: $\tau (=RC)$	Comments
Process	Experimentally: $\tau = -\frac{1}{m}$		
Charging			
Discharging			

7.11 Discussion

7.12 Resources

For further understanding students may go through the following resources:

- **Fundamental of Physics (10th Edition):** Capacitor (Chapter 25, page 717-721), RC circuit (Chapter 27, page 788-791).
- **Video Links:**
 - https://www.youtube.com/watch?v=f_MZNsEqyQw
 - [\(4\) 22 - Circuits - Time constant of an RC circuit - YouTube](#)